TIP STUDY

The GLM Procedure

Class Level Information		
Class	**Levels**	**Values**
TTIP	4	1 2 3 4
COUPON	4	1 2 3 4

Number of Observations Read	16
Number of Observations Used	15

TIP STUDY

The GLM Procedure

Dependent Variable: DEPRESS COUPON DEPRESSION

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	1.19511111	0.19918519	25.61	<.0001
Error	8	0.06222222	0.00777778		
Corrected Total	14	1.25733333			

R-Square	Coeff Var	Root MSE	DEPRESS Mean
0.950513	0.917389	0.088192	9.613333

Source	DF	Type I SS	Mean Square	F Value	Pr > F
TTIP	3	0.40566667	0.13522222	17.39	0.0007
COUPON	3	0.78944444	0.26314815	33.83	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
TTIP	3	0.39527778	0.13175926	16.94	0.0008
COUPON	3	0.78944444	0.26314815	33.83	<.0001

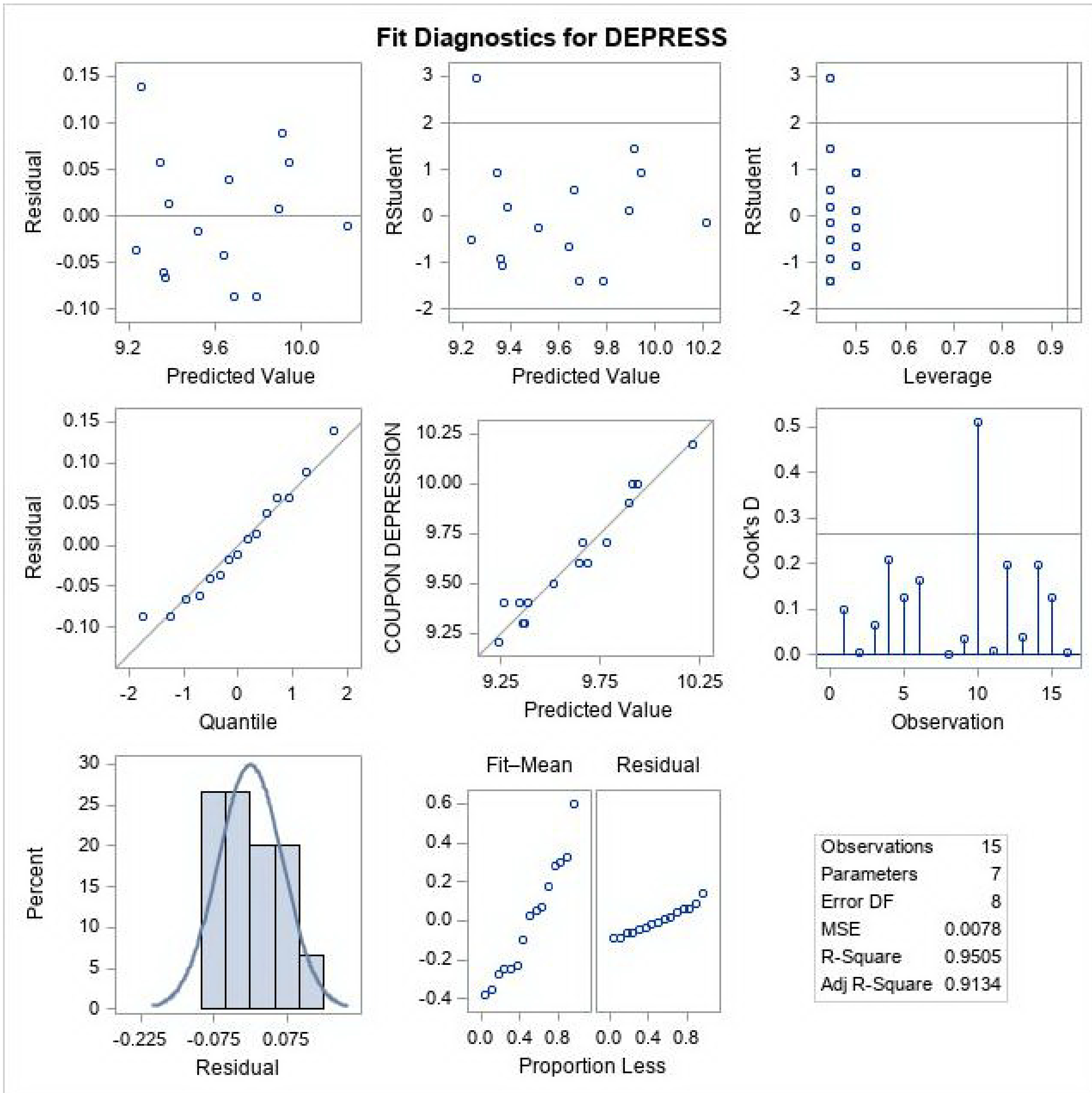
TIP STUDY

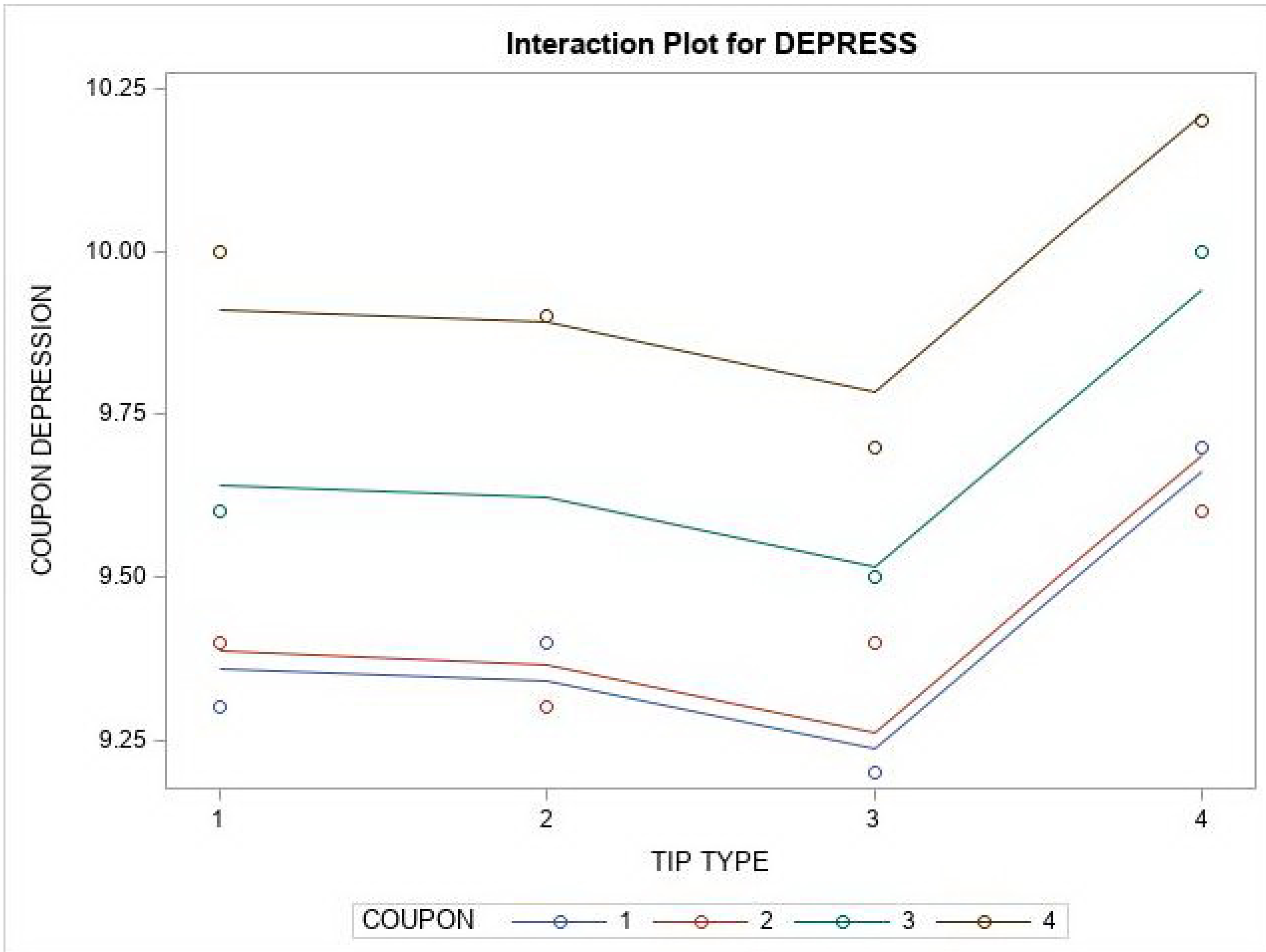
The GLM Procedure

Observation		Observed	Predicted	Residual
1		9.30000000	9.36111111	-0.06111111
2		9.40000000	9.38611111	0.01388889
3		9.60000000	9.64166667	-0.04166667
4		10.00000000	9.91111111	0.08888889
5		9.40000000	9.34166667	0.05833333
6		9.30000000	9.36666667	-0.06666667
7	*	.	9.62222222	.
8		9.90000000	9.89166667	0.00833333
9		9.20000000	9.23611111	-0.03611111
10		9.40000000	9.26111111	0.13888889
11		9.50000000	9.51666667	-0.01666667
12		9.70000000	9.78611111	-0.08611111
13		9.70000000	9.66111111	0.03888889
14		9.60000000	9.68611111	-0.08611111
15		10.00000000	9.94166667	0.05833333
16		10.20000000	10.21111111	-0.01111111

* Observation was not used in this analysis

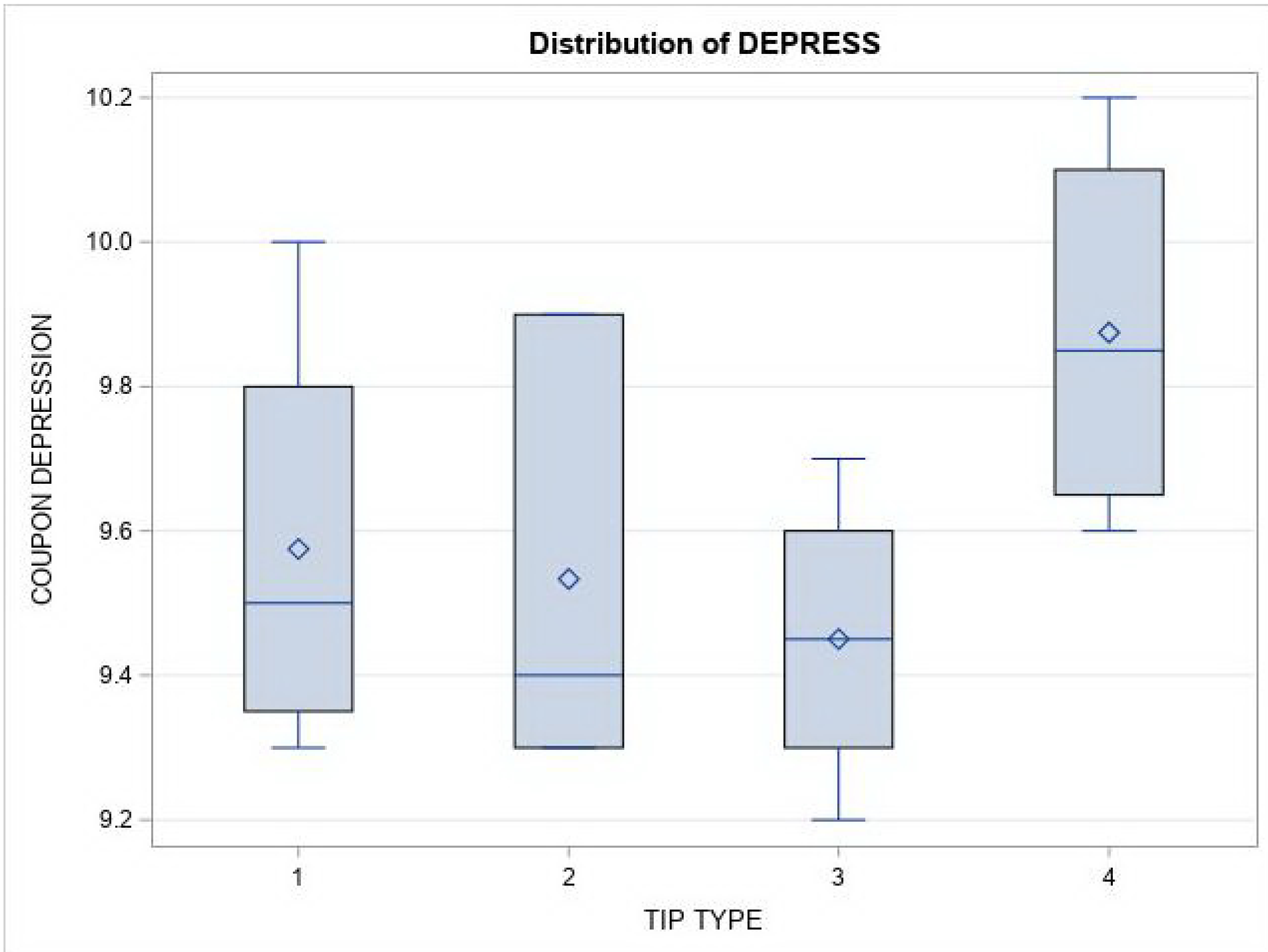
Sum of Residuals	-0.00000000
Sum of Squared Residuals	0.06222222
Sum of Squared Residuals - Error SS	-0.00000000
First Order Autocorrelation	-0.36892361
Durbin-Watson D	2.67584325


Fit Diagnostics for DEPRESS
Residual
Predicted Value
RStudent
Predicted Value
RStudent
Leverage
Residual
Quantile
COUPON DEPRESSION
Predicted Value
Cook's D
Observation
Percent
Residual
Fit–Mean
Residual
Proportion Less
Observations 15
Parameters 7
Error DF 8
MSE 0.0078
R-Square 0.9505
Adj R-Square 0.9134


Interaction Plot for DEPRESS
10.25
10.00
9.75
9.50
9.25
COUPON DEPRESSION
1
2
3
4
TIP TYPE
COUPON 1 2 3 4

TIP STUDY

The GLM Procedure



TIP STUDY

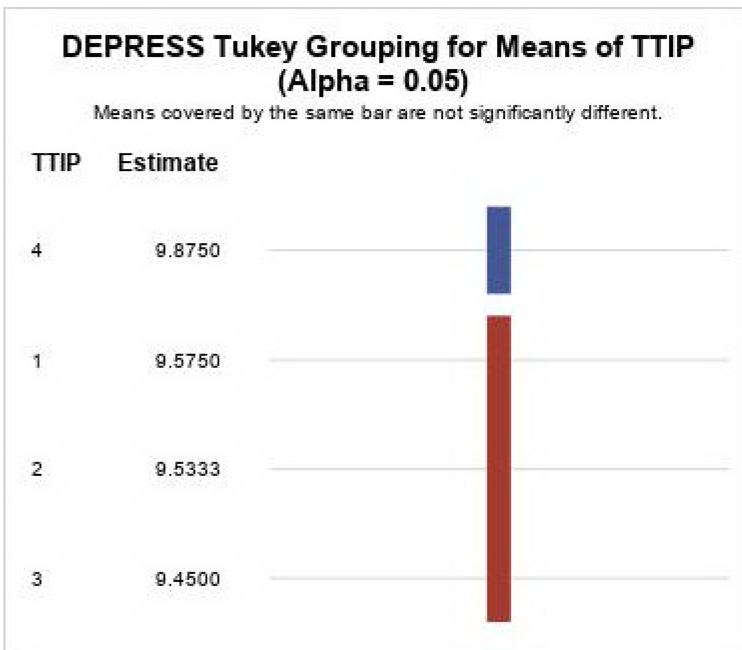
The GLM Procedure

Tukey's Studentized Range (HSD) Test for DEPRESS

Note: This test controls the Type I experimentwise error rate, but it generally has a higher Type II error rate than REGWQ.

Alpha	0.05
Error Degrees of Freedom	8
Error Mean Square	0.007778
Critical Value of Studentized Range	4.52877
Minimum Significant Difference	0.2079
Harmonic Mean of Cell Sizes	3.692308

Note: Cell sizes are not equal.



TIP STUDY

The UNIVARIATE Procedure
Variable: RESID

Moments			
N	15	Sum Weights	15
Mean	0	Sum Observations	0
Std Deviation	0.06666667	Variance	0.00444444
Skewness	0.53228022	Kurtosis	-0.3840287
Uncorrected SS	0.06222222	Corrected SS	0.06222222
Coeff Variation	.	Std Error Mean	0.01721326

Basic Statistical Measures			
Location		Variability	
Mean	0.00000	Std Deviation	0.06667
Median	-0.01111	Variance	0.00444
Mode	-0.08611	Range	0.22500
		Interquartile Range	0.11944

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	0	Pr > |t|	1.0000
Sign	M	-0.5	Pr >= |M|	1.0000
Signed Rank	S	-4	Pr >= |S|	0.8346

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.954866	Pr < W	0.6040
Kolmogorov-Smirnov	D	0.105976	Pr > D	>0.1500
Cramer-von Mises	W-Sq	0.030519	Pr > W-Sq	>0.2500
Anderson-Darling	A-Sq	0.228158	Pr > A-Sq	>0.2500

Quantiles (Definition 5)	
Level	Quantile
100% Max	0.1388889
99%	0.1388889

95%	0.1388889
90%	0.0888889
75% Q3	0.0583333
50% Median	-0.0111111
25% Q1	-0.0611111
10%	-0.0861111
5%	-0.0861111
1%	-0.0861111
0% Min	-0.0861111

Extreme Observations

Lowest		Highest	
Value	**Obs**	**Value**	**Obs**
-0.0861111	14	0.0388889	13
-0.0861111	12	0.0583333	5
-0.0666667	6	0.0583333	15
-0.0611111	1	0.0888889	4
-0.0416667	3	0.1388889	10

Missing Values

Missing Value	Count	Percent Of	
		All Obs	**Missing Obs**
.	1	6.25	100.00